الجمهورية اللبنانية
وزارة التربية والتعليم العالي
المديرية العامة للتعليم المهني والتقني

الامتحانات الرسمية لشهادة الامتياز الفني
لدورة عام ٢٠٢٣الاولى

الاسم والشهرة:----------
رقم المرشح :----------

الاختصاص والرمز : الالكترونيك (٢٠٢٠٣)
المادة : الالكترونيك الرقمي
المستندات المسموح بها : لا شيء

المعدل: (١٨)
المدة : ساعتان

الإجابات على "كراس الإجابة"

A- Choisir la réponse correcte: (1/2 pt / réponse correcte)

1- Si le registre DDRA = $(1111\ 1111)_2$ alors PORTA sera initialisé en:
- a- Huit entrées
- b- Huit sorties
- c- 4 entrées et 4 sorties
- d- Aucune des réponses

2- Si le registre DDRA = $(0000\ 1111)_B$ alors PORTA sera initialisé:
- a- Huit entrées
- b- Huit sorties
- c- 4 entrées et 4 sorties
- d- Aucune des réponses

3- Le résultat si l'instruction [ANDB #$FF] avec (B) = $(FF)_{16}$ est:
- a- (B) = $(FF)_{16}$
- b- (B) = $(AA)_{16}$
- c- (B) = $(FA)_{16}$
- d- (B) = $(00)_{16}$

4- Le résultat si l'instruction [ADDB #$FF] avec B = $(FF)_{16}$ est:
- a- (B) = $(FF)_{16}$
- b- (B) = $(AA)_{16}$
- c- (B) = $(FA)_{16}$
- d- (B) = Aucune des réponses

5- Pour obtenir la différence $(180 - 70)_{10}$ avec B = $(180)_{10}$, on écrit l'instruction suivante:
- a- SUBB #$10
- b- SUBB #$70
- c- SUBB #$0070
- d- Aucune des réponses

6- Pour allumer, LED1 et LED2 connectées par leurs anodes aux broches PB1 et PB2 du PORTB, on écrit LDB #$06 et:
- a- STA DDRBB
- b- STA ORA
- c- STA ORB
- d- STB ORB

7- Le résultat de l'instruction [EORA #$AA] avec (A) = $(FF)_{16}$ est:
- a- (A) = $(AA)_{16}$
- b- (A) = $(FF)_{16}$
- c- (A) = $(55)_{16}$
- d-(A) = $(5A)_{16}$

8- Que contient la mémoire $0040 à la fin de ce programme :
- a- ($0040) = $(0A)_{16}$
- b- ($0040) = $(0B)_{16}$
- c- (0040$) = $(AA)_{16}$
- d- (0040$) = $(10)_{16}$

```
LDB#S02
LDA#$05
MUL
STB $0040
END
```

9- Le résultat de l'instruction [ORA #$AA] avec A = $FF sera:
- a- (A) = $FF
- b- (A) = $FA
- c- (A) = $00
- d- (A) = $AA

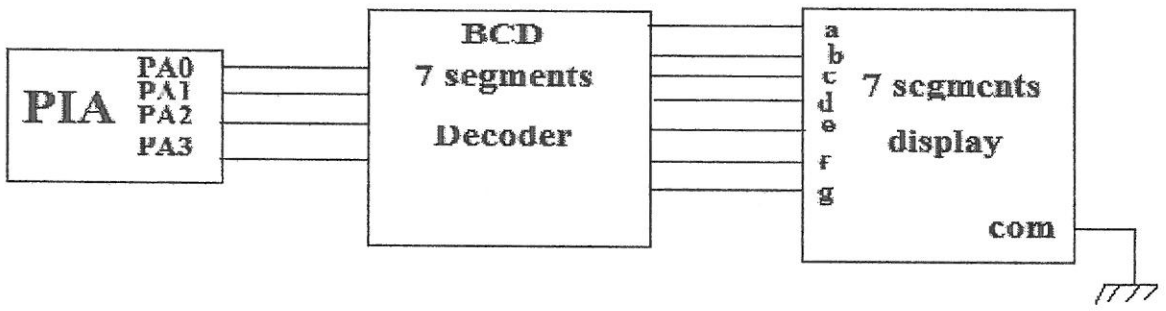
10- Le résultat de l'instruction [LDA $40] avec ($40) = $(1F)_H$ est:
- a- (A) = $(40)_H$
- b- (A) = $(80)_H$
- c- (A) = $(1F)_H$
- d- (A) = $(4F)_H$

B - Traiter 3 questions seulement : (5 pts / question)

I- Ecrire le programme qui fait sortir séquentiellement sur le port B d'un PIA6821 connecté au MP6809, les octets du tableau suivant.

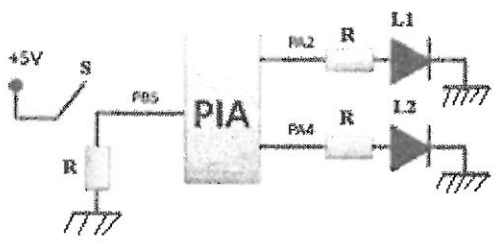
PB7	PB6	PB5	PB4	PB3	PB2	PB1	PB0
0	0	0	0	0	0	0	1
0	0	0	0	0	0	1	0
0	0	0	0	0	1	0	0
0	0	0	0	1	0	0	0
0	0	0	1	0	0	0	0
0	0	1	0	0	0	0	0
0	1	0	0	0	0	0	0
1	0	0	0	0	0	0	0

II- En vous référant au circuit ci-dessous, écrivez le programme pour construire un compteur de 0 à F.



III- Ecrire en langage 6809 le programme qui exécute ce qui suit:

1- S est OUVERT → L1 ON L2 OFF
2- S est FERMÉ → L1 OFF L2 ON



IV- Ecrire en langage 6809 le programme, qui multiplie deux nombres entiers positifs, le premier se trouvant dans l'adresse $0020 et le second est une constante égale à $04, puis stocke le résultat dans l'adresse $0050

Instruction set of 6809

ADCA, ADCB, ADDA, ADDB, ADDD, ANDA, ANDB, ANDCC, ASLA, ASLB, ASL, ASRA, ASRB, ASR, BITA, BITB, CLRA, CLRB, CLR, CMPA, CMPB, CMPD, CMPU, CMPS, CMPX, CMPY, COMA, COMB, COM, CWAI, DAA, DECA, DECB, DEC, EORA, EORB, EXG, INCA, INCB, INC, JMP, JSR, LDA, LDB, LDD, LDS, LDU, LDX, LDY, LEAS, LEAU, LEAX, LEAY, LSLA, LSLB, LSL, LSRA, LSRB, LSR, MUL, NEGA, NEGB, NEG, NOP, ORA, ORB, ORCC, PSHS, PSHU, PULS, PULU, ROLA, ROLB, ROL, RORA, RORB, ROR, RTI, RTS, SBCA, SBCB, SEX, STA, STB, STD, STS, STU, STX, STY, SUBA, SUBB, SUBD, SWI, SWI2, SWI3, SYNC, TFR, TSTA, TSTB, TST.

PIA:

ORA et DDRA = $8000	**CRA = $8001**
ORB et DDRB = $8002	**CRB = $8003**

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المعدل: (١٨)
المدة : ساعتان

الإجابات على "كراس الإجابة"

A- Choose the correct answer : (1/2 pt / correct answer)

1- If the register DDRA = $(11111111)_2$ then PORTA will be initialized as :
- a- Eight inputs
- b- Eight outputs
- c- Four inputs and four outputs
- d- None of the answers

2- If the register DDRA = $(0000\ 1111)_2$ then PORTA will be initialized as :
- a- Eight inputs
- b- Eight outputs
- c- Four inputs and four outputs
- d- None of the answers

3- The result of the instruction [ANDB #$FF] with (B) = $(FF)_{16}$ is:
- a- (B) = $(FF)_{16}$
- b- (B) = $(AA)_{16}$
- c- (B) = $(FA)_{16}$
- d- (B) = $(00)_{16}$

4- The result of the instruction [ADDB #$FF] with (B) = $(FF)_{16}$ is:
- a- (B) = $(FF)_{16}$
- b- (B) = $(AA)_{16}$
- c- (B) = $(FA)_{16}$
- d- (B) = $(00)_{16}$

5- To achieve the difference $(180 - 70)_{10}$ with (B) = $(180)_{10}$, we write the following instruction:
- a- SUBB #$10
- b- SUBB #$70
- c- SUBB #$0070
- d- None of the answers

6- To turn ON, LED1 and LED2 connected by their anodes to pins PB1 and PB2 of PORTB, we write LDB #$06 and:
- a- STA DDRBB
- b- STA ORA
- c- STA ORB
- d- STB ORB

7- The result of the instruction [EORA #$AA] with (A) = $(FF)_{16}$ is :
- a- (A) = $(AA)_{16}$
- b- (A) = $(FF)_{16}$
- c- (A) = $(55)_{16}$
- d- (A) = $(5A)_{16}$

8- What is the content of the memory $0040 at the end of this program:
- a- ($0040) = $(0A)_{16}$
- b- ($0040) = $(0B)_{16}$
- c- ($0040) = $(AA)_{16}$
- d- ($0040) = $(10)_{16}$

```
LDB#$02
LDA#$05
MUL
STB $0040
END
```

9- The result of the instruction [ORA #$AA] with (A) = $FF is :
- **a-** (A) = $FF
- **b-** (A) = $FA
- **c-** (A) = $00
- **d-** (A) = $AA

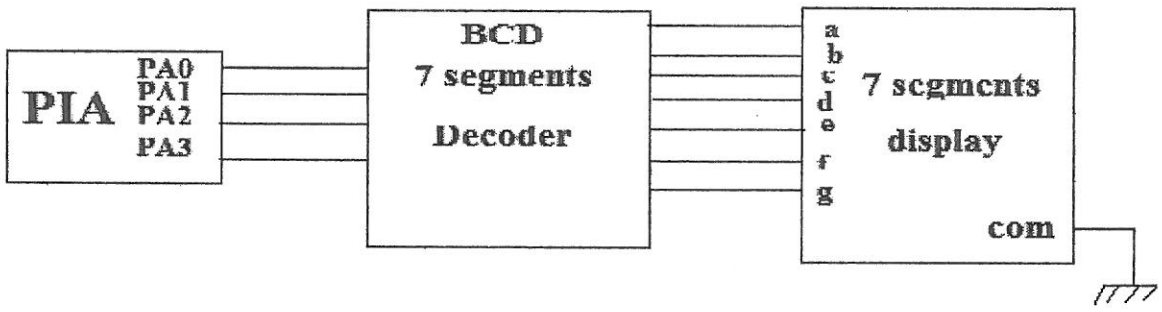
10- The result of the instruction [LDA $40] with $40 = $(1F)_H$ will be :
- a- (A) = $(40)_H$
- b- (A) = $(80)_H$
- c- (A) = $(1F)_H$
- d- (A) = $(4F)_H$

B - Treat 3 questions only: (5 pts / question)

I- Write the program that outputs sequentially at port B of a PIA6821 connected to the MP6809, the bytes of the following table.

PB7	PB6	PB5	PB4	PB3	PB2	PB1	PB0
0	0	0	0	0	0	0	1
0	0	0	0	0	0	1	0
0	0	0	0	0	1	0	0
0	0	0	0	1	0	0	0
0	0	0	1	0	0	0	0
0	0	1	0	0	0	0	0
0	1	0	0	0	0	0	0
1	0	0	0	0	0	0	0

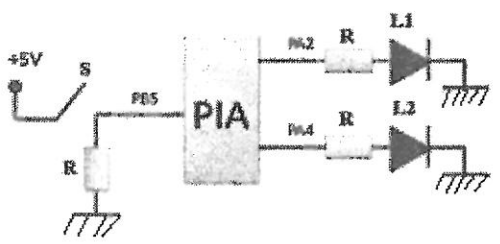
II- By referring to the below circuit, write the program to build a counter from 0 to 9.



III- Write in 6809 language the program to execute does the following:

1- S is OPEN → L1 ON L2 OFF

2- S is CLOSED → L1 OFF L2 ON



IV- Write in 6809 language the program that multiplies two integer and positive numbers, the first being in the address $0020 and the second is a constantequal to $04 then stores the result in the address $0050

ANNEXE / APPENDIX

Instruction set of 6809

ADCA, ADCB, ADDA, ADDB, ADDD, ANDA, ANDB, ANDCC, ASLA, ASLB, ASL, ASRA, ASRB, ASR, BITA, BITB, CLRA, CLRB, CLR, CMPA, CMPB, CMPD, CMPU, CMPS, CMPX, CMPY, COMA, COMB, COM, CWAI, DAA, DECA, DECB, DEC, EORA, EORB, EXG, INCA, INCB, INC, JMP, JSR, LDA, LDB, LDD, LDS, LDU, LDX, LDY, LEAS, LEAU, LEAX, LEAY, LSLA, LSLB, LSL, LSRA, LSRB, LSR, MUL, NEGA, NEGB, NEG, NOP, ORA, ORB, ORCC, PSHS, PSHU, PULS, PULU, ROLA, ROLB, ROL, RORA, RORB, ROR, RTI, RTS, SBCA, SBCB, SEX, STA, STB, STD, STS, STU, STX, STY, SUBA, SUBB, SUBD, SWI, SWI2, SWI3, SYNC, TFR, TSTA, TSTB, TST.

PIA:

ORA et DDRA = $8000	**CRA = $8001**
ORB et DDRB = $8002	**CRB = $8003**